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What is varroa mite, and why is it a threat to crop pollination?

Varroa mite, *Varroa destructor*, is a small mite, around 1 mm in diameter, that parasitises species of bees in the genus *Apis*, such as *Apis cerana*, the Asian honey bee, and *Apis mellifera*, the species that includes the European honey bee. In the 20th century two strains of *V. destructor* moved hosts from the natural host, the Asian honey bee, to the European honey bee. The K strain of *Varroa destructor* (K stands for Korea, the country in which the strain is thought to have originated) has spread world-wide and is regarded as the major challenge facing beekeeping internationally.

A strain of *V. jacobsoni* that can parasitise *A. mellifera* was discovered in Papua New Guinea in 2008, but its distribution remains restricted (Roberts et al., 2014).

Untreated hives of European honey bees infested with varroa are likely to die within 3 to 4 years. In other countries, populations of feral (wild) European honey bees not managed by beekeepers, have fallen by more than 90 per cent following the establishment of varroa. The decrease in the populations of European honey bees reduces the amount of pollination services that are available to crops (Rosenkranz et al., 2010).

What is pollination and what types of pollination are there?

Pollination is the process in which pollen is transferred from the male part of the flower to the female part of the flower, thereby fertilising the ovule. The fertilised ovule grows into a seed which is contained in the fruiting part of the plant. Many of the grains, fruits and vegetables that we eat are the fruiting part of the plant which either contains or wholly comprises the seeds. Without pollination, the fruiting part of the plant does not form.

Flowers are pollinated in different ways. Wind is sufficient to transfer the pollen from the male to the female parts of the flower in some plants. These plants are said to be 'wind pollinated'. In other plants, the pollen is transferred without the need for any transfer agent. These plants are said to be 'passive self-pollinated'. In other plants an animal, often an insect, is required to transfer the pollen from the male to the female flower parts. These plants are said to be 'animal pollinated'. Most plants are pollinated through a combination of the different mechanisms-i.e. some wind pollination and some animal pollination.

Are European honey bees better than other insects at pollinating crops?

Bees are better than any other group of animals at pollinating plants and have evolved with life cycles intimately connected with plants. European honey bees are the best known and most widely spread group of bees that have been domesticated to pollinate the crops eaten by humans. European honey bees can be reared and maintained domestically and deployed for large scale pollination, making them the most widely used pollinator of many crops. Although many other insects are more efficient

pollinators of particular crops, because European honey bees are used at a high density they can pollinate a diverse range of crops.

Are all crops pollinated by insects?

No. The bulk of energy in our diets and feedstocks for animals comes from crops, such as wheat, corn and rice, that are exclusively wind or passively self-pollinated and don't require insect pollination (Table 1).

However, insect pollination is essential for fruit-set in other crops and these crops would have a yield reduction of more than 90 per cent without insect pollination. Insect pollination also contributes to the quality of produce, such as fruit size. Crops, such as broccoli or onions, don't require insect pollination to form the part of the crop we eat, but insect pollination is important in generating seed to plant these crops. The range of dependence of crops to insect pollination is given in Table 1.

Table 1. The level of biological dependence of crops to insect pollination (Klein et al., 1997)

Essential	Great	Modest	Little	None
	Apple			
				Sugar cane
	Mango			
Kiwifruit		Cotton		Corn/Maize
	Blackberries and related berries	Coffee	Capsicum	Wheat
Passionfruit			Tomato	
Macadamia	Cherries	Faba bean	Kidney bean	Rice
Watermelon	Plums			Barley
		Soya bean		
Rockmelon	Avocado		Peanut	Sorghum
		Sunflower		
Pumpkin, squash and zucchini	Almonds	Papaya		Chickpea
		Chestnut		
	Canola			Table and wine grapes
	Cucumber			

Essential = pollinators essential for most varieties (production reduction by 90 % or more comparing experiments with and without animal pollination)

Great = animal pollinators are strongly need (40-90 percent reduction)

Modest = animal pollinators are clearly beneficial (10-40 percentreduction)

Little = Some evidence suggests that animal pollinators are beneficial (0-10 percent reduction)

None = no production increase with animal mediated pollination

How much do feral (wild) European honey bees contribute to the pollination of Australian crops?

The European honey bee was introduced to Australia around 1822 and since then has spread widely across the country. It is uncertain how much feral European honey bees contribute to the pollination of Australian crops and how much is contributed by native insects.

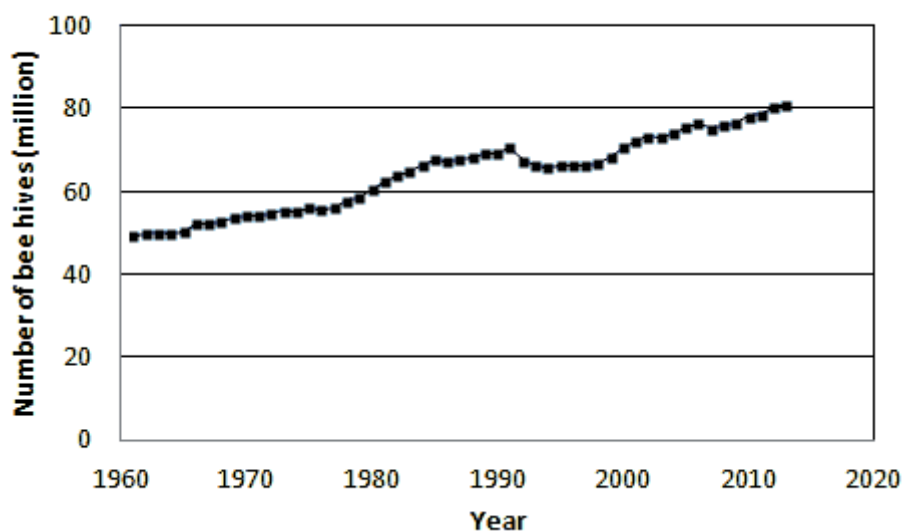
Which other countries have varroa mite?

All countries with significant beekeeping industries have varroa mite, except Australia. There are some countries in central Africa and some Pacific Island nations that do not have varroa mite, but these countries lack significant beekeeping industries.

What effect has varroa had on the number of managed bee hives in other countries?

Since 1961 the reported global stock of commercial managed honey bee hives has increased by approximately 64 per cent (Figure 1).

Figure 1. The number of managed honey bee hives in the world from 1961-2013 (FAO Stat, 2015).

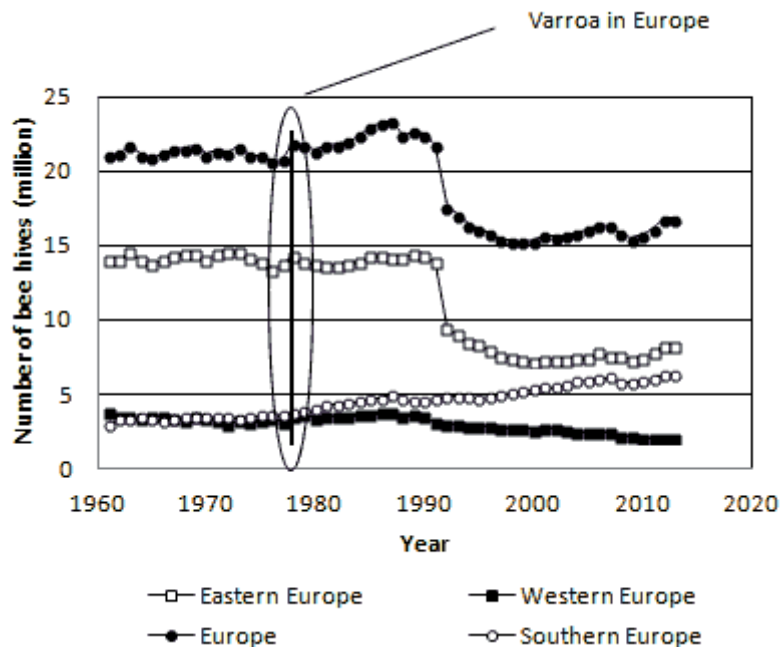


The number of bee hives in a particular country depends on a range of social and economic factors affecting the beekeeping industry in that country; varroa is only one of these factors (vanEngelsdorp and Meixner, 2010).

Varroa had no perceptible effect on the number of hives reported in Europe (Figure 2). The number of honey bee hives in Europe declined sharply in the early 1990s coinciding with the end of communism, and the end of state support for beekeepers, in the previously communist bloc countries of Eastern Europe (Moritz et al., 2010). The number of hives reported in Western European countries declined

slightly over the same period of time and the number of hives reported in Southern European countries increased.

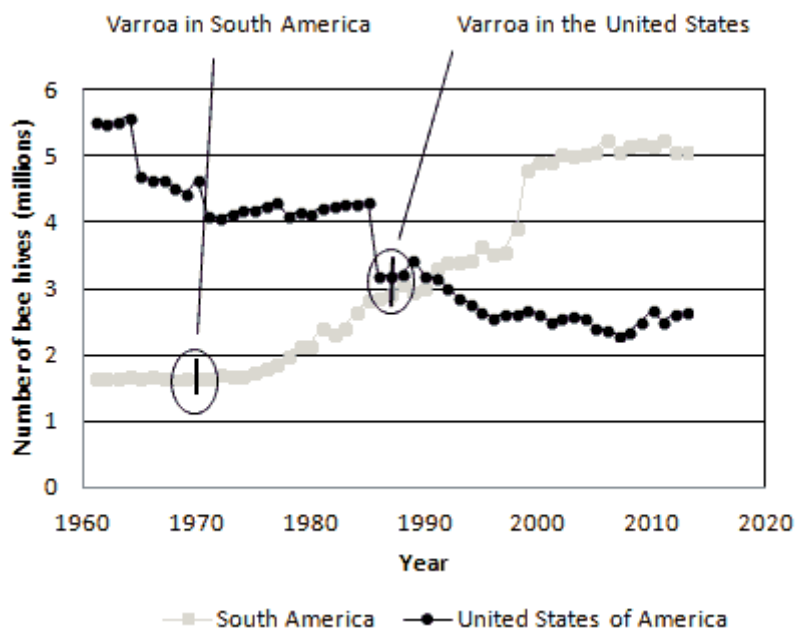
Figure 2. The number of managed hives in the whole of Europe and Eastern, Western and Southern Europe from 1961-2013 (FAO Stat, 2015).



Rising incomes in rural areas, a declining rural workforce and the need to treat for varroa has led hobby beekeeping to become less popular in some European countries (Potts et al., 2010).

In the United States the number of managed hives has declined steadily since the late 1940s, around 40 years before varroa established there. This decline reflects declining terms of trade for United States beekeepers as the result of competition with lower-cost honey producing countries in South America. In contrast, due to their competitive advantage, the number of hives in South America has grown steadily since the mid-1970s, despite varroa already being established there (Figure 3). However, the J strain of *V. destructor* in South America is less damaging than the K strain of *V. destructor* in the United States.

Figure 3. The number of managed honey bee hives in the United States and South American countries from 1961-2013 (FAO Stat, 2015).



What border biosecurity arrangements are in place to prevent varroa getting into Australia?

The most likely entry pathway for varroa into Australia is associated with its host, the European honey bee, on a vessel or conveyance of international origin or with bees brought into Australia illegally. The Australian Government Department of Agriculture's work to prevent the entry of bees from foreign countries focuses on three possible entry pathways:

- vessels or conveyances arriving into the country
- cargo and containers on board those vessels
- the movement of people, goods and vessels through the Torres Strait.

All vessels arriving in Australia must undergo quarantine clearance. Quarantine clearance may be in the form of assessment of documentation, physical inspection, targeted/random surveillance or a combination of these processes.

All vessels arriving into Australia are required to submit a Quarantine Pre Arrival Report (QPAR) to the department between 96 and 12 hours prior to arrival. A question to which all masters must answer is "Have any insects, including bees, been discovered on board during this voyage?", if so then a description of the insect and its location will be required.

Where bees have been reported or detected on a vessel prior to its arrival in port:

- an department entomologist is consulted for advice and directions on appropriate management of the bees on the incoming vessel and cargo, which can include meeting the vessel offshore
- the vessel is advised of any special requirements on entry into the port by adding them to the Approval to Berth form provided to the vessel in response to the QPAR
- cargo is prohibited from being discharged from the vessel until the vessel has been cleared of the bee risk

- the vessel is inspected thoroughly with extra attention applied to locations where bees might inhabit.

The department conducts surveillance on vessels and port precincts if bee activity is reported or discovered by or on arriving vessels. One of the department's entomologists is contacted for treatment directions.

Appropriate communication material and contact information for stevedores and other port occupants is available to assist in timely reporting and management of bee activities.

If bees are discovered within cargo or on containers, they are ordered for treatment according to the requirements prescribed by an entomologist.

The Northern Australian Quarantine Strategy (NAQS) monitors the movement of all vessels, people and goods into Torres Strait islands. A major part of that program is the monitoring of the movement of traditional and non-traditional vessels between Papua New Guinea, the Torres Strait Protected Zone, the Special Quarantine Zones and the mainland across the Torres Strait. NAQS scientists also monitor the environs of these zones for evidence of incursions of exotic pests and disease.

What arrangements are in place to eradicate varroa should it find its way into Australia?

A response to an incursion of varroa (to contain and, if possible eradicate varroa) is coordinated by the Australian Chief Plant Protection Office of the Australian Government Department of Agriculture. Government authorities will follow the predetermined response plans laid out in the Emergency Plant Pest Response Deed (EPPRD). The EPPRD is an agreement between governments and industries as to actions to occur in responding to pests of biosecurity concern.

PLANTPLAN provides nationally consistent guidelines for response procedures, outlining the phases of an incursion (investigation, alert, operational and stand down), as well as the key roles and responsibilities of industry and government during each of these phases. It is updated regularly to incorporate new information or address gaps identified by the outcomes of pest of biosecurity concern incident reviews.

What are the chances of eradicating varroa mite if it's found in Australia?

The chances of eradicating an incursion depend on the nature of the incursion, in particular how widespread it is when it is discovered. A widespread incursion is unlikely to be eradicable.

How does the establishment of *Apis cerana* (Asian honey bee) affect the ability of the Australian honey bee industry to manage the future establishment of varroa?

In 2007, Asian honey bees were detected in Cairns, Queensland, and a national eradication program was undertaken. In 2011, a decision was made that it was no longer technically feasible to achieve

eradication of the bees. On 20 May 2011 the then Minister for Agriculture, Fisheries and Forestry, Senator the Hon Joe Ludwig, announced funding of \$2 million to support a national pilot program aimed at transitioning to containment and management of Asian honey bees.

At this time it is not certain what effect the Asian honey bee will have on the Australian honey bee industry. However, the presence of *A. cerana* is not likely to have a direct impact on the spread of varroa throughout Australia. The strains of *Varroa destructor* that exist on *A. mellifera* are not known to move from *A. mellifera* onto *A. cerana* hence *A. cerana* will not act as a vector of this parasite whereas feral *A. mellifera* will. Additionally the movement of strains of *V. jacobsoni* from *A. cerana* onto *A. mellifera* is a rare occurrence—it appears that this occurs only when there has been long term exposure of both of the species of *Apis* as well as a strain of *V. jacobsoni* that has the characteristics of being able to survive on *A. mellifera*.

Now Australia has Asian honey bee, is the threat of varroa mite more likely?

The Asian honey bee now in Australia is not carrying varroa.

The greatest threat to the European honey bee are the strains of *V. destructor* and *V. jacobsoni* that parasitise the European honey bee.

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Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/bees/honeybees-faqs>